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Enhancing self-directed learning with custom GPT AI facilitation among medical students: A randomized controlled trial

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ABSTRACT

Objective: This study aims to assess the impact of LearnGuide, a specialized ChatGPT tool designed to support self-directed learning among medical students.

Materials and methods: In this 14-week randomized controlled trial (ClinicalTrials.gov NCT06276049), 103 medical students were assigned to either an intervention group, which received 12 weeks of problem-based training with LearnGuide support, or a control group, which received identical training without AI assistance. Primary and secondary outcomes, including Self-Directed Learning Scale scores at 6 and 12 weeks, Cornell Critical Thinking Test Level Z scores, and Global Flow Scores, were evaluated with a 14-week follow-up. Mann-Whitney U tests were used for statistical comparisons between the groups.

Results: At 6 weeks, the intervention group showed a marginally higher median Self-Directed Learning Scale score, which further improved by 12 weeks (4.15 [95% CI, 0.82 to 7.48]; $p = 0.01$) and was sustained at the 14-week follow-up. Additionally, this group demonstrated notable improvements in the Cornell Critical Thinking Test Score at 12 weeks (7.11 [95% CI, 4.50 to 9.72]; $p < 0.001$), which persisted into the 14-week follow-up. The group also experienced enhancements in the Global Flow Score from 6 weeks, maintaining superiority over the control group through 12 weeks.

Conclusions: LearnGuide significantly enhanced self-directed learning, critical thinking, and flow experiences in medical students, highlighting the crucial role of AI tools in advancing medical education.

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KEYWORDS
Artificial intelligence; self-directed learning; critical thinking; problem-based learning; ChatGPT

1. Introduction

Self-directed learning (SDL) is delineated by the capacity and motivation of learners to establish and pursue their individual learning objectives, as well as to evaluate their progress and outcomes [1]. SDL emerges as a cornerstone for the cultivation of critical thinking and problem-solving abilities, indispensable for patient care. This process facilitates students' ability to stay abreast of medical innovations, a crucial aspect of a physician's continuous professional development.

Throughout the COVID-19 pandemic, the educational landscape experienced a significant shift from traditional in-person instruction to online learning platforms [2]. This transition, marked by a decline in attendance at conventional lectures and an increased reliance on diverse external educational resources [3], has highlighted the crucial role of SDL in enhancing the effectiveness of online education and ensuring student adaptability and success in the modern digital learning environment [4]. As this educational paradigm continues to evolve, the cultivation of SDL skills among students has become more essential than ever.

ChatGPT, a powerful large language model, has demonstrated its proficiency in passing medical exams in various languages, highlighting its potential in the medical field [5,6]. Its increasing popularity among medical educators and students has led to improvements in the creation of multiple-choice questions [7], teaching evaluations [8], and

Practice points

- LearnGuide, a ChatGPT-based AI tool, significantly enhances self-directed learning and critical thinking skills in medical education.
- The benefits of LearnGuide, such as improved learning and critical thinking, are sustained over a 14-week period, showing the lasting impact of AI-facilitated education.
- LearnGuide increases student engagement and flow, leading to more immersive and effective learning experiences.
- Strategically integrating AI tools like LearnGuide within educational frameworks such as PBL enhances SDL without fostering over-reliance on these tools among students.
- LearnGuide potentially optimizes self-directed learning by reducing extraneous cognitive load and customizing tasks to align with students' developmental stages.

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academic writing support [9]. Although ChatGPT has shown value in educational settings by simplifying various tasks, its true impact on enhancing students' learning capabilities, particularly SDL skills, remains uncertain.

Integrating ChatGPT to develop an innovative teaching system that enhances students' SDL shows great promise. This online, student-centered learning approach has the potential to reduce reliance on educator guidance typically required to foster autonomy in traditional teaching methods [10–12]. However, the extensive use of ChatGPT in learning has raised concerns among some educators that its use could lead to overreliance [13,14], potentially hindering the development of essential learning skills. It is suggested that an effective learning process depends not just on the tools used, but more importantly on how learners actively interact with and explore the educational content. A comprehensive evaluation is urgently needed to understand how ChatGPT can be integrated into current educational systems to enhance students' learning capabilities, particularly SDL skills.

To bridge this gap, we developed LearnGuide, a customized version of ChatGPT specifically designed to enhance SDL skills. This innovative study rigorously evaluates the integration of LearnGuide with problem-based learning (PBL) on the improvement of SDL, critical thinking skills, and flow experiences. Our findings highlight the significant role of ChatGPT as an 'artificial intelligence (AI) facilitator' in enhancing students' SDL skills, providing a promising model for integrating AI into medical education.

2. Methods

This 14-week, open-label, randomized controlled trial, initiated in November 2023 at the Second Xiangya Hospital of Central South University, comprised a 12-week intervention period followed by a 2-week follow-up, marks the first investigation into the efficacy of custom GPT in enhancing SDL among students (Trial Registration ClinicalTrials.gov Identifier: NCT06276049, Trial Protocol in Supplement 1).

2.1. Developing 'LearnGuide': A custom GPT for SDL enhancement

Custom GPTs, recently introduced as custom versions of ChatGPT, can be effortlessly created for specific tasks or topics. These can be distributed widely *via* shared links and GPT stores. Utilizing the GPT builder, we developed 'LearnGuide'(https://chatgpt.com/g/g-oZ8zdPaKp-learnguide), a bespoke GPT designed to serve as an 'AI facilitator' for SDL (eFigure 1, eFigure 2, eAppendix 1 in Supplement 2). Through interactive dialogue with users, it enhances engagement and provides support across their SDL journey (eAppendix 2, 3 in Supplement 2).

In the process of facilitating SDL among students, key components such as self-assessment, goal setting, active learning, and evaluation are integral, all underpinned by instructor-led guidance. Accordingly, we have equipped LearnGuide with six functional modules: Linguistic Mastery Assistant, Innovative Reasoning Boost, Knowledge and Resource Facilitation, Empathetic Interactive Learning Support, Learning Recap, and Self-Assessment Guide (eTable 1 in Supplement 2 for details). These modules are designed to enhance LearnGuide's capacity in supporting learning and evaluation processes with greater precision and proactive assistance, in alignment with the SDL instructional framework.

2.2. Participants

Undergraduate and postgraduate students studying at the Second Xiangya Hospital, Central South University, with access to ChatGPT were eligible to participate in the study. Recruitment occurred from November to December 2023, utilizing WeChat, China's most commonly used social and messaging application. Participation in the study was voluntary. The exclusion criterion was defined as utilization of large language models for learning assistance more than once weekly for a period exceeding 2 months.

2.3. Description of intervention

The intervention consisted of a LearnGuide utilization guide course and a 12-week integrated training program employing LearnGuide. During the initial 2-hour group session, instructional staff introduced and conducted a hands-on demonstration of LearnGuide. This included acquainting students with the fundamental principles of ChatGPT, such as crafting high-quality prompts for efficient interaction. Furthermore, the instructional team highlighted LearnGuide's comprehensive function as an 'AI facilitator' within the SDL framework. Subsequent to the introductory session, instructors assigned a range of integrated PBL training tasks. Throughout the following 12-week SDL phase with LearnGuide, students were mandated to undertake at least one task weekly from the provided selection. This component of the intervention afforded students the chance to seek advice from mentors regarding inquiries related to LearnGuide.

2.4. Study groups

103 undergraduate and postgraduate participants specializing in clinical medicine at the Second Xiangya Hospital were evenly distributed and allocated to either the intervention or control groups using a computer-generated 1:1 randomization algorithm. The intervention group received a LearnGuide-supported integrated training program. In contrast, the control group completed 12 weeks of integrated training tasks, scheduled once weekly, without the support of LearnGuide. The flowchart is shown in the Figure 1.

2.5. Data collection and study measures

At baseline, as well as at 6, 12 (intervention phase), and 14 weeks (follow-up phase), participants were required to complete the Self-Directed Learning Scale (SDLS) and the Cornell Critical Thinking Test (CCTT) Level Z following a comprehensive overview of the questionnaire protocol. This procedure was implemented to assess their SDL and critical thinking skills. Participants were required to complete the Flow Short Scale within 30 min following the initial, 6-week, and 12-week training tasks. The Flow Short Scale is employed to assess the general propensity to experience flow states and comprises ten items rated on a 7-point scale.

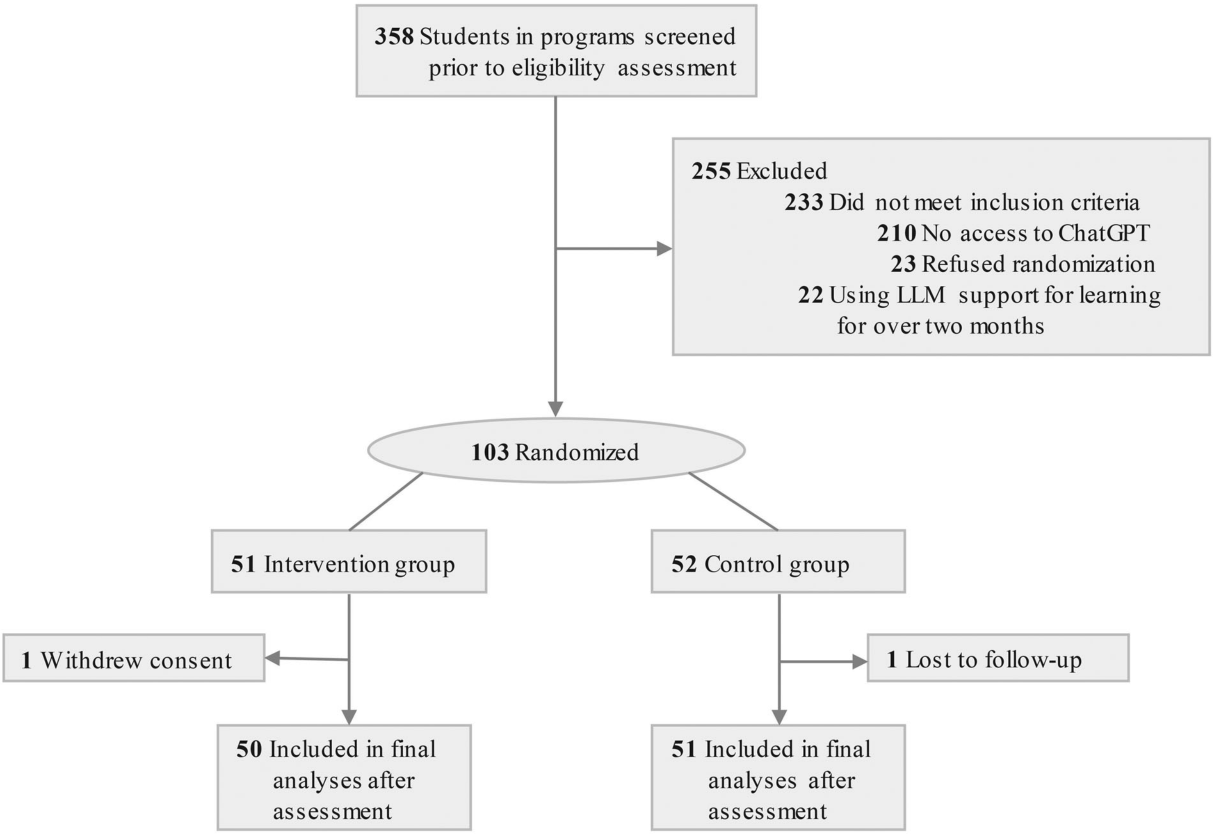

Figure 1. Enrollment of participants and study Flow.

The SDLS, utilizing a five-point Likert scale across ten items, assesses learning autonomy in adolescents and adults. Anchored in Brockett's framework, it quantifies learners' independent learning inclination, exhibiting robust internal consistency and validity [15,16]. The CCTT Level Z measures cognitive abilities in students beyond 11th grade with 52 multiple-choice questions across six domains based on the Ennis and Millman Theory, including deductive and inductive reasoning, semantic analysis, credibility assessment, experimental planning, and assumption identification [17]. The Global Flow Score (GFS) is calculated by averaging the scores of the ten items on the Flow State Scale [18].

2.6. Sample size

The sample size was estimated assuming a target effect size of 0.60 [19], reflecting a moderate to large impact of our intervention on the SDLS scores. Calculations indicated that 45 participants per group would be required to achieve 80% power at a 0.05 significance level for two-sided tests. To account for a potential 10% attrition rate, we adjusted our total sample size to 100 participants, with 50 in each group.

2.7. Statistical analysis

We applied SPSS statistical software version 23 (IBM Statistics) to perform statistical analyses for this trial. Continuous variables were presented as means with standard deviations, and categorical variables as frequencies and percentages. Given that GFS, CCTT, and other scores derived from Likert scale responses do not follow a normal distribution, we represented these data using medians and interquartile ranges (IQRs), which better describe non-normally distributed data. The Mann-Whitney U test was used for comparing the central tendencies of these non-normally distributed, independent samples. Independent samples t-test were used where normality and homogeneity assumptions were satisfied (e.g. age and entrance English score), otherwise Mann-Whitney U tests were performed. The χ^2 test is used for categorical covariates (e.g. gender and educational level).

2.8. Qualitative analysis

Feedback from participants using LearnGuide, collected at week 12 through open-ended questions, offered valuable perspectives on its application scenarios and fulfillment of their needs during the intervention period. This feedback underwent thematic analysis based on Braun and Clarke's methodology [20], leading to a thematic map that displays the findings quantitatively.

3. Results

3.1. Participant

In November 2023, we screened 358 undergraduate and postgraduate students for eligibility. Out of these, 255 were excluded due to lack of access to ChatGPT, refusal to participate, or previous engagement with Large Language Model (LLM) support for learning exceeding 2 months. Consequently, 103 students met all inclusion criteria for our study and were subsequently randomized into two groups: 52 students in the intervention group and 51 students in the control group. During the study, there was one loss to follow-up in the control group and one withdrawal from the intervention group (Figure 1).

Table 1. Participant demographic characteristics.

Characteristic/baseline scores	Control group ($n = 52$)	Intervention group ($n = 51$)	*P*-value[a]
Age, Mean (SD), y	24.8 (2.6)	25.0 (2.0)	0.73
Gender, No. (%)			
Male	30 (58)	27 (53)	
Female	22 (42)	24 (47)	0.77
Education level, No. (%)			
Undergraduate	22 (42)	21 (41)	
Postgraduate	30 (58)	30 (59)	1.0
Ethnicity, No. (%)			
Han	44 (85)	48 (94)	
Non-Han	8 (5.5)	3 (6)	0.21
Entrance English score (SD)[b]	84.6 (5.6)	85.9 (4.6)	0.22

[a]Statistical analysis methods include independent samples t-test and χ^2 test.
[b]Scores were standardized to percentages by dividing obtained marks by the maximum possible and multiplying by 100. This ensures comparability across educational levels.

Table 2. Baseline SDLS, CCTS, and GFS scores of participants.

Characteristic/baseline scores	Control group ($n = 52$)	Intervention group ($n = 51$)	Intervention vs control, mean difference (95% CI)	*P*-valuea
Baseline SDLS Score (median [IQR]) (range, 10-50)	27.0 (24.0 to 30.5)	28.0 (26.5 to 30.0)	1.3 (-0.2 to 2.8)	0.10
Baseline CCTS score (median [IQR]) (range, 0-52)	25.0 (22.0 to 28.5)	24.0 (21.5 to 28.0)	1.1 (-3.1 to 0.9)	0.23
Baseline GFS (median [IQR]) (range 1-7)[a]	3.90 (3.30 to 4.40)	4.20 (3.75 to 4.75)	0.24 (-0.02 to 0.50)	0.06

SDLS: Self-Directed Learning Scale; CCTS: Cornell Critical Thinking Test; GFS: Global Flow Score.
[a]Statistical analysis conducted using Mann-Whitney U test.
[b]GFS is calculated using the Flow Short Scale, with participants responding to items that are aggregated into an overall average score from 10 questions.

Table 3. Outcome variables at intervention and follow-up.

Outcome variable	Median (IQR) Control group ($n = 51$)	Intervention group ($n = 50$)	Intervention vs control, mean difference (95% CI)	*P*-value[a]
SDLS score, (range, 10-50)				
6 wk	28.0 (23.5 to 31.0)	31.0 (26.0 to 37.8)	3.1 (0.0 to 6.2)	0.05
12 wk	29.0 (24.0 to 35.0)	33.5 (26.3 to 42.0)	4.2 (0.8 to 7.5)	0.01
14 wk	28.4 (24.3 to 32.3)	34.4 (30.4 to 39.2)	5.9 (3.2 to 8.5)	<.001
CCTT score, (range, 0-52)				
6 wk	24.0 (20.0 to 29.5)	25.0 (21.3 to 31.8)	1.3 (-1.6 to 4.2)	0.27
12 wk	27.0 (22.5 to 29.0)	33.5 (29.0 to 39.5)	7.1 (4.5 to 9.7)	<0.001
14 wk	26.0 (23.0 to 30.0)	33.0 (29.0 to 36.0)	6.0 (4.0 to 8.0)	<0.001
GFS (range 1-7)[b]				
6 wk	4.20 (3.75 to 4.45)	4.65 (4.30 to 5.00)	0.59 (0.38 to 0.80)	<0.001
12 wk	4.50 (4.05 to 4.85)	5.30 (4.93 to 5.78)	0.81 (0.53 to 1.10)	<0.001

SDLS: Self-Directed Learning Scale; CCTS: Cornell Critical Thinking Test; GFS: Global Flow Score.
[a]Statistical analysis conducted using Mann-Whitney U test.
[b]The Global Flow Score is derived by dividing the Flow State Scale score by 10, which represents the number of items.

3.2. Demographic Characteristics and baseline scores

103 Students (43 undergraduate and 60 postgraduate) participated in the study, including 30 females (57.7%). The mean ages (SD) of the control and intervention groups were 24.8 (2.6) years and 25.0 (2.0) years, respectively. The predominant ethnicity in both groups was Han, with comparable ratios of Han to non-Han members in each group. No significant differences were observed in terms of gender, ethnicity, or entrance English score between the groups (Table 1). Baseline assessments of the SDLS and the CCTS revealed no significant differences between the groups. However, post-initial training, the intervention group exhibited a marginally higher GFS compared to the control group (mean difference, 0.24 [95% CI, −0.02 to 0.50)], p=.06) (Table 2).

3.3. Main outcomes

At the 12-week conclusion of the study, both groups exhibited improvements in their SDLS scores, CCTT scores, and GFS compared to baseline (eTable 2 in supplement 2). After 6 weeks, the intervention group demonstrated a marginally higher median SDLS score (median [IQR]) compared to the control group (31.0 [26.0-37.8] vs. 28.0 [23.5-31.0]; mean difference, 3.08 [95% CI, 0.01 to 6.15]; p=0.05). By 12 weeks, this difference had increased, with the intervention group showing further improvement (4.15 [95% CI, 0.82 to 7.48]; $p = 0.01$). Notably, at the 14-week follow-up, the intervention group maintained a significant advantage over the control group (5.87 [95% CI, 3.22 to 8.51]; $p < 0.001$). For the CCTT score, at 6 weeks, the difference between the groups was not statistically significant. However, by 12 weeks, the intervention group exhibited a significant improvement (33.5 [29.0-39.5] vs. 27.0 [22.5-29.0]; 7.11 [95% CI, 4.50 to 9.72]; $p < 0.001$), which remained significant at the 14-week follow-up (6.00 [95% CI, 3.97 to 8.02]; $p < 0.001$). In terms of Global Flow Scores, the intervention group had a significantly higher score at 6 weeks compared to the control group (4.65 [4.3-5.0] vs. 4.20 [3.75-4.45]; 0.59 [95% CI, 0.38 to 0.80]; $p < 0.001$). This improvement was maintained at 12 weeks (0.81 [95% CI, 0.53 to 1.10]; $p < 0.001$) (Table 3).

3.4. Qualitative analysis of students Feedback

Feedback thematic analysis ($N = 52$) identified 'Positive Experiences,' 'Neutral Experiences,' and 'Negative Experiences'

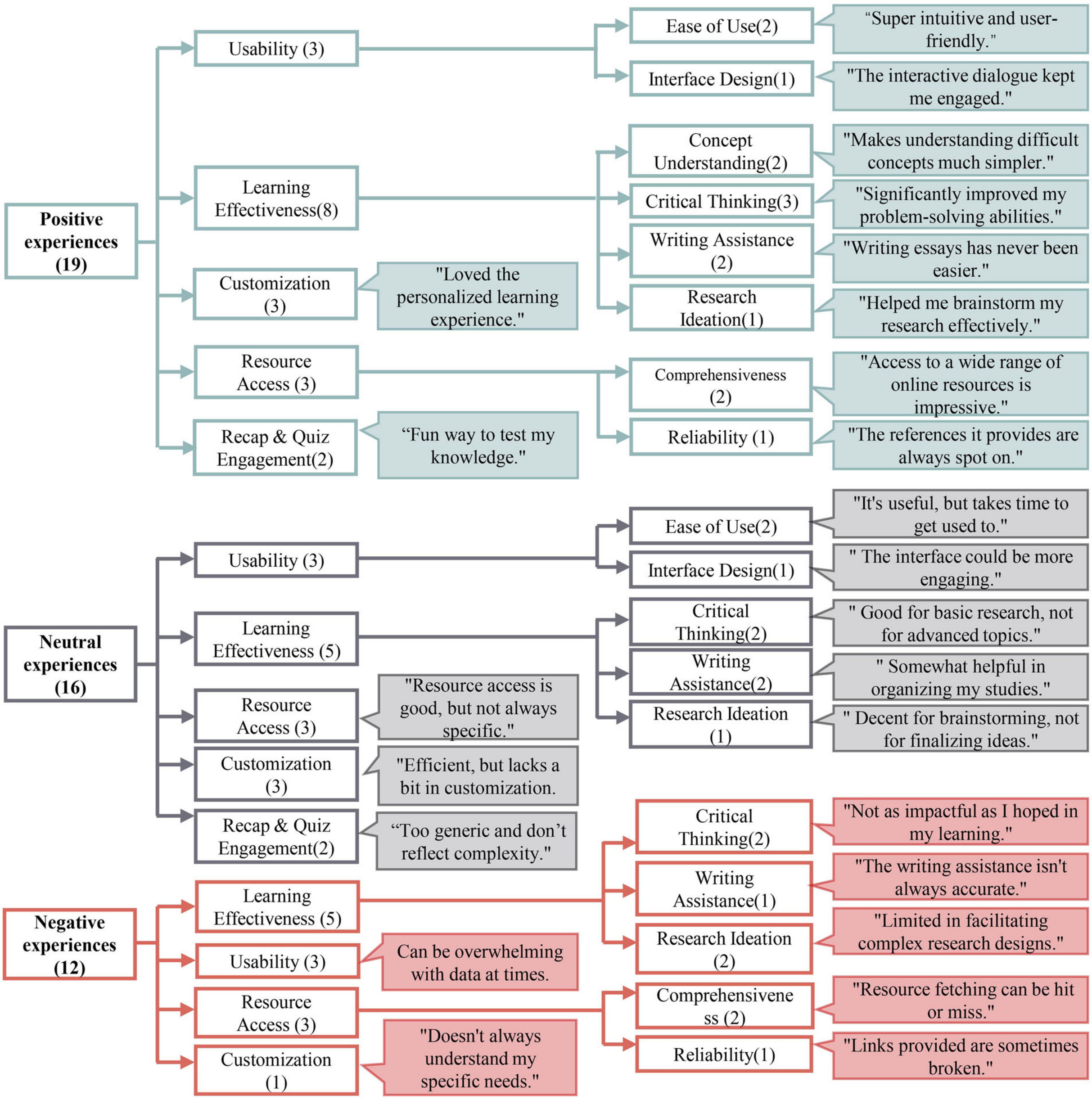

Figure 2. Exploring the impact of LearnGuide through thematic analysis of student Feedback and examples.
Thematic analysis of intervention group feedback identified primary themes and subthemes, with representative feedback examples shown; the number of participants who identified each is indicated in brackets.

as primary themes, with subthemes: 'usability,' 'learning effectiveness,' 'resource access,' 'customization,' and 'comparative analysis.' Positive feedback praised LearnGuide's user-friendly design and educational value, emphasizing its intuitive interface and interactive learning experiences. The majority of user experiences were either positive or neutral, and correlational analysis indicated that these experience categories have a positive, albeit weak, correlation with scores on the SDLS and the CCTT (eTable 3 in supplement 2). The 'learning effectiveness' subtheme was noted for making complex topics accessible and encouraging analytical thinking. 'Resource access' received accolades for comprehensive, reliable content. Neutral feedback highlighted LearnGuide's reliability and customization potential, suggesting improvements and noting occasional content relevance issues. Negative comments addressed technical issues and comparative limitations, like response lags, navigation difficulties, and inadequacies in complex academic content analysis, especially for in-depth medical literature (Figure 2). The findings suggest LearnGuide benefits SDL development but also reveal areas for enhancement, including personalization, technical reliability, and content depth, to meet diverse student needs.

4. Discussion

The implementation of LearnGuide in PBL task-based training has resulted in statistically significant enhancements in SDLS scores. This confirms our hypothesis that strategically embedding specially designed ChatGPT can boost the development of SDL skills in medical education. Notably, SDL improvements were evident at 6 weeks and further increased at 12 weeks. Following a 2-week follow-up period, the intervention group's SDLS scores remained significantly higher, indicating the durability of enhanced SDL skills. This suggests that prolonged use of LearnGuide might further enhance SDL, meriting further research into LearnGuide and similar large language models in SDL contexts.

The study also highlights significant improvements in GFS, attributed to LearnGuide's interactive dialogue, which promotes real-time engagement and sustained focus, thereby enhancing the flow experience from the study's outset [21,22]. Feedback from learners has praised LearnGuide for its user-friendly interface, its efficacy in clarifying complex concepts, and providing comprehensive, reliable resources. This optimization of learning experiences and performance fosters an enhanced flow experience, which in turn naturally boosts motivation, focus, and autonomy—key components in supporting effective SDL [4].

Furthermore, LearnGuide effectively reduces extraneous cognitive load by utilizing worked examples to elucidate complex or unfamiliar concepts, specifically tailored to the learner's existing knowledge. Simultaneously, it modifies intrinsic load by simplifying tasks appropriate to the developmental stage of the learner, ensuring the context remains intact. High levels of extraneous cognitive load can divert student attention and deplete cognitive resources, thus impeding learning. On the other hand, excessively high intrinsic cognitive load can cause frustration and diminish motivation, whereas insufficiently challenging tasks may not fully engage the learner's capabilities [23].

The significant increase in scores on the CCTT during the intervention and follow-up phases underscores the significant role of LearnGuide in bolstering critical thinking skills, crucial for medical practice. This outcome aligns with prior research indicating SDL as a key intermediary in enhancing the nexus between critical thinking and problem-solving skills [24]. Considering the pivotal role of critical thinking dispositions in applying analytical skills for problem evaluation and solving [25], future research should explore shifts in these dispositions among participants.

In fostering SDL skills, educators are tasked with creating supportive and resource-rich learning environments. Traditionally, this entails guided learning under instructor oversight, fostering learner independence within a structured setting [26–28]. Integration of LearnGuide revolutionizes this by encouraging students to proactively explore and start a self-driven learning, enabling them to efficiently search for interdisciplinary information, critically analyze concepts, and perform self-evaluations. This makes LearnGuide a potential pioneer in promoting SDL without direct teacher involvement, proving particularly advantageous in online learning scenarios where face-to-face interaction is minimal.

However, it is critical to acknowledge that ChatGPT is not a panacea in the context of SDL; indeed, its application may sometimes prove detrimental. For instance, reliance on LearnGuide solely for completing test assignments or for paraphrasing academic papers to circumvent plagiarism detection does not contribute to the enhancement of SDL. It's more about how learners use tools like LearnGuide for SDL, not just the tools themselves. Effective SDL comes from engaging with the content and how it's used [29]. Therefore, integrating tools like LearnGuide into existing educational frameworks beyond PBL has become a critical topic that educators must carefully consider.

Despite LearnGuide being a custom GPT developed by our team, it remains accessible to educational institutions and students *via* a public link. Furthermore, educational institutions and students are encouraged to consult the creation guidelines for this custom GPT, leveraging their insights to develop their own refined version of 'LearnGuide.' The availability of such resources is pivotal to the worldwide distribution and adoption of our methodology, aiming to enhance SDL.

A considerable number of participants were excluded due to unavailable ChatGPT access, indicating those included might already have a predisposition towards ChatGPT-enhanced learning. This scenario suggests LearnGuide's potentially greater benefit to these 'better-prepared' students, thereby introducing selection bias. Such bias could tilt our findings towards those naturally inclined towards SDL, hence narrowing the relevance of our conclusions to a broader spectrum of medical students lacking ChatGPT access or familiarity.

Currently, free ChatGPT accounts can access custom GPTs, including LearnGuide. However, the development of custom GPTs still requires a paid premium subscription. Consequently, simply using LearnGuide may not present a significant barrier to students and educational institutions, even in lower-income regions. Nevertheless, the accessibility of ChatGPT is restricted in several countries, including China, which limits its global adoption and effectiveness in enhancing education worldwide. Fortunately, other open-source and free LLMs, such as the recently launched ChatGPT 4o, are being developed. These alternatives have the potential to offer broader access to AI-supported learning for more students. These emerging alternatives could not only improve user experiences but also potentially enhance the facilitation of SDL on a broader scale.

Non-native English speakers using LearnGuide or ChatGPT-4 for semantic-based information retrieval in their native languages [30], especially when searching for materials in English and interdisciplinary studies, may find it particularly beneficial. This advantage may not be as pronounced for native English speakers [31]. Although this study included graduate students with varying levels of English proficiency, future research incorporating native English speakers is necessary to determine LearnGuide's effectiveness in enhancing SDL among this demographic.

Another potential limitation is LearnGuide's adaptability across various medical specialties. Its effectiveness in enhancing SDL could differ by subject, particularly where practical, hands-on experience is essential. This suggests a need for future studies to assess LearnGuide's utility and effectiveness across a wider spectrum of medical fields, to fully grasp its capabilities and constraints.

In conclusion, our study shows that LearnGuide, a ChatGPT-based AI, significantly enhances SDL in medical students when integrated with PBL, leading to noticeable improvements in SDL skills, critical thinking, and flow experiences. These findings highlight LearnGuide's effectiveness as a student-oriented educational tool in medical education.

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Authors' contributions

Wang Shalong: data collection and writing of the manuscript; Zuo Yi, providing suggestions; Liu Ganglei, Zhou Jinyu, Zheng Yanwen, Zhang

Zequn and Yuan Lianwen: data collection and organization; Zou Bin: data analysis; Ren Feng: revision of the manuscript, organization and providing suggestions.

Ethics statement

The study received ethical approval from the Ethics Committee of the Second Xiangya Hospital, Central South University. Informed consent was secured from all participants before their engagement in the study.

Disclosure statement

The funding sources had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication. No potential conflict of interest was reported by the authors.

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